

UNDERSEA LAUNCH

Nuclear-powered submarines are equipped with SLBMs and nuclear-tipped cruise missiles.

VULNERABILITY: None. It is extremely difficult to detect a submarine in the open ocean. Deters an adversary from launching a crippling first strike that can knock out nuclear assets. ATV is equipped with a solid-fuelled, 750-km range SLBM. DRDO is already working on equipping India's SSBN fleet with the extended range Agni-III, a three-stage SLBM with a range of 5,000 km.



man the submarine, is being trained at an indigenously-developed simulator in the School for Advanced Underwater Warfare (SAUW) at the naval base in Vizag. Hands-on training will be done on the *INS Chakra*, a 12,000-tonne Akula-II class nuclear-powered attack submarine being taken on a 10-year lease from Russia next year. SBC in Vizag is to become the assembly line for three ARVs, costing a little over Rs 3,000 crore each or the cost of a 37,000 tonne indigenous aircraft carrier built at the Cochin Shipyard.

Larsen and Toubro (L&T) has begun building the hull of the second ATV at its facility in Hazira, to be inducted into the navy by 2012. The SSBN fleet will be housed on the east coast at a new naval base in Rambilli, a few kilometres south of Visakhapatnam, where nearly 3,000 acre of land has been acquired for India's first strategic base, to be manned entirely by military personnel. Unlike the narrow single channel in Visakhapatnam, it will offer the nuclear fleet direct access into the sea. The first phase of the project, costing approximately Rs 1,500 crore, will be ready by 2011.

Why has the project taken so long? For a country that built only two conventional submarines of the German-designed HDW Type 1500 class in the

early '90s, building a nuclear submarine was the ultimate challenge: a DRDO official sees the learning curve to be the equivalent of a scooter mechanic building a Mercedes. The key challenge, however, was not in designing or fabricating the hull, but the reactor and containment vessel, which consumes one-tenth (nearly 600 tonne) of the vessel's total displacement. The hydrodynamics of a vessel with one-tenth of its weight concentrated in one place is a formidable naval engineering challenge, but miniaturising a nuclear reactor the size of a football field to fit inside an 8m enclosure is an even bigger hurdle. This was among the reasons for the decade-long delay in the project. The nuclear reactor in a submarine generates heat to convert water into saturated steam to turn the submarine's turbines. Unlike an oil-fired boiler, it does not require air to operate. All other parts of the subma-

PROJECT INDIA: A submarine from Russia will train the ATV crew



rine are the same as any steam-powered turbine plant's. The reactor operates on uranium enriched to nearly 45 per cent (uranium used in civilian nuclear reactors is less than 5 per cent and bombs use uranium enriched to over 90 per cent).

In 1998, L&T began fabricating the hull of ATV but the struggle with the reactor continued. After BARC designs failed, India bought reactor designs from Russia. By 2004 the reactor had been built, tested on land at the IGCAR and had gone critical. Its modest size, around 6,000 tonne (the Ohio class SSBN in the movie *Crimson Tide* weighs over 14,000 tonne), has led experts to call it a "baby boomer". While the present project ends at three units, defence officials have not ruled out building larger submarines on the basis of national strategic imperatives. These have changed since the conception of the project.

The plan, until late '80s, was to build an SSN—a fast-moving deep-diving nuclear-powered attack submarine, which would hunt surface ships. Around the time India leased a Charlie-I class nuclear-powered attack submarine from the Soviet Union, it had already veered towards building a submarine carrying ballistic missiles. The hull design was lengthened and the SSN quietly transformed into an SSBN.

There are, however, some key challenges to be overcome. ATV's SLBMs have a range of only 750 km, a big leap from its start of 250 km a decade ago, but still smaller than the SLBMs deployed by the Big Five, which boast ranges in excess of 5,000 km. DRDO is working on fielding a submarine launched variant of the 5,000-km Agni III missile, which will give the submarine true striking power and flexibility.

Scientists believe the submarine's present reactor output of around 80 MW is limited because it imposes operational restrictions on the submarine's speed and will mean that the reactor will have to function near peak power at most times. The reactor would also need constant refuelling—a fairly expensive process where the hull is cut open and the nuclear cores replaced every decade. For the moment, however, the immediate challenge lies in successfully sending the submarine out to the sea. ■

"It is a DRDO project and somewhere near completion, it will be in the water in two years."

ADMIRAL SURESH MEHTA, CHIEF OF THE NAVAL STAFF

